Kaustubh Sridhar

Research Scientist @ Google Deepmind

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Education

Aug 2019 - May 2025 **University of Pennsylvania**, *Philadelphia, PA.*
PhD Candidate, **Electrical and Systems Engineering**, GPA: 3.94/4.
ASSET and PRECISE Center
Thesis Title: Training Adaptive and Sample-Efficient Autonomous Agents
Thesis Committee: Prof Dinesh Jayaraman, Prof Insup Lee, Prof George Pappas, Prof Nikolai Matni, Prof Dorsa Sadigh

Jul 2015 - May 2019 **Indian Institute of Technology Bombay**, *Mumbai, India.*
Bachelor Of Technology (with Honors) In Aerospace Engineering, *GPA: 9.07/10.*
Minor in Systems and Control Engineering Class Rank 2.

Work Experience

May 2025 - Present **Google Deepmind**, *Senior Research Scientist*, New York, NY.

- **Genie 4** and **Genie Next**
  - Pre-trained and post-trained **Genie 4**, a realtime interactive omni-modal (omni-input/output) model, and the successor to Genie 3 in the Project Genie experience.
  - Driving video-to-video pre-training on **Genie Next**, an highly-efficient version of Genie 4.
- **Gemini Omni** to **World Action Models**
  - Pre-trained and post-trained Gemini Omni, an omni-input/output model, on robotics and human egocentric datasets to create a dexterous World Action Model (WAM) policy.
- **SIMA 2** [2025C, 2025B, videos]
  - Post-trained Gemini flash-sized vision-language-models (VLMs) to predict keyboard and mouse actions to complete any long-horizon task in virtual worlds.
  - Spearheaded SIMA's autonomous self-improvement in unseen worlds, to human-level performance, with a Gemini-based reward function and a Gemini-based task-setter.
- **RoboGenie** and **Gemini Robotics 2** [2026A, videos]:
  - Post-trained the Genie 3 world model on robotics datasets to create a world model for humanoid and tabletop (aloha and franka) robots.
  - RoboGenie enabled planning with Gemini Robotics 2 vision-language-action models (VLAs), doubling their success rate with more test-time compute on long-horizon/long-tail tasks in dexterous manipulation and whole-body control.

Aug 2019 - May 2025 **University of Pennsylvania**, *PhD Candidate*, Philadelphia, PA.
Advised by Prof Insup Lee.
Closely collaborated with Prof Dinesh Jayaraman, Prof Oleg Sokolsky.

- Post-trained a VLA to add in-context adaptability to boost its performance on new robotic tasks with unseen objects, motions, and scenes [2025A, videos].
- Pretrained a generalist agent that can generalize to unseen robotics and game-playing environments via retrieval-augmentation & in-context learning [2024A, videos].
- Strengthened imitation learning with any neural network – diffusion models, transformers, or MLPs, via a novel semi-parametric model class called the MCNN [2023B, videos].
- Created a tool for guaranteed conformance of generative models to constraints [2023A, gifs].
- Boosted deep hierarchical RL sample-efficiency by two-orders-of-magnitude [2022D, videos].
- Enhanced adversarial robustness of NN's with guarantees [2022A].
- Developed out-of-distribution detectors with guarantees [2022B] that run in real-time [2021B].

May - Aug 2023 **Amazon Web Services (AWS) AI Labs**, *Applied Scientist Intern*, Santa Clara, CA.
Hosts: Dr. Abishek Sankararaman, Dr. Vikram Nathan, Dr. Murali Narayanaswamy

- Improved generalization in offline RL by incorporating transformer model based forecasts in conservative Q learning; applied to cloud resource allocation problems.

May - Aug 2022 **Amazon Web Services (AWS) AI Labs**, *Applied Scientist Intern*, Santa Clara, CA.
Hosts: Dr. Abishek Sankararaman, Dr. Murali Narayanaswamy
- Accelerated datacenter resource allocation by combining model-free RL with mixed integer linear programs [Preprint 2022C].

May - Aug 2021 **Argo AI (Ford & VW's Self-Driving Partner)**, *Systems Research Intern*, Dearborn, MI.
Product Security and Sensor Functional Safety Team
- Built threat models for object detection and segmentation models on autonomous vehicles.

Research Interests

I am interested in creating adaptive generalist agents for the digital and physical worlds. Towards this goal, I have worked on multimodal generative models, deep reinforcement and imitation learning, and in-context learning.

Selected Publications and Preprints

[2026A] Gemini Robotics 2 brings whole body intelligence to robots
Gemini Robotics 2 team including **Kaustubh Sridhar**
➥ Google Deepmind Blogpost.

[2025C] SIMA 2: A Generalist Embodied Agent for Virtual Worlds
SIMA 2 team including **Kaustubh Sridhar**
➥ Technical Report, arXiv:2512.04797.

[2025B] SIMA 2: An Agent that Plays, Reasons, and Learns With You in Virtual 3D Worlds
SIMA 2 team including **Kaustubh Sridhar**
➥ Google Deepmind Blogpost.

[2025A] RICL: Adding In-Context Adaptability to Pre-Trained Vision-Language-Action Models
Kaustubh Sridhar, Souradeep Dutta, Dinesh Jayaraman, Insup Lee
➥ Conference on Robot Learning (CoRL) 2025 (Acceptance rate: 35%).

[2024A] REGENT: A Retrieval-Augmented Generalist Agent That Can Act In-Context In New Environments
Kaustubh Sridhar, Souradeep Dutta, Dinesh Jayaraman, Insup Lee
➥ International Conference on Learning Representations (**ICLR**) **2025**,
➥ **Oral presentation** at ICLR 2025, **top 1.8% of 11672 submissions**,
➥ **NeurIPS 2024** workshops on Adaptive Foundation Models and Open World Agents.

[2023B] Memory-Consistent Neural Networks for Imitation Learning
Kaustubh Sridhar, Souradeep Dutta, Dinesh Jayaraman, James Weimer, Insup Lee
➥ International Conference on Learning Representations (**ICLR**) **2024** (Acceptance rate: 31%).

[2023A] Guaranteed Conformance of Neurosymbolic (World) Models to Natural Constraints
Kaustubh Sridhar, Souradeep Dutta, James Weimer, Insup Lee
➥ **ICLR 2023** workshop on Neurosymbolic Generative Models,
➥ Conference on Learning For Dynamics and Control (**L4DC**) **2023**.

[2022D] Exploring with Sticky Mittens: Reinforcement Learning with Expert Interventions via Option Templates
S. Dutta*, **K. Sridhar***, O. Bastani, E. Dobriban, J. Weimer, I. Lee, J. Parish-Morris
➥ Conference on Robot Learning (**CoRL**) **2022** (Acceptance rate: 39%).

[Preprint 2022C] Predict-and-Critic: Accelerated End-to-End Predictive Control for Cloud Computing through Reinforcement Learning
Kaustubh Sridhar, Vikramank Singh†, Murali Narayanaswamy†, Abishek Sankararaman†
➥ Under review (†**Amazon AWS AI Labs**).

[2022B] CODiT: Conformal Out-of-distribution Detection in Time-series Data
Ramneet Kaur, **Kaustubh Sridhar**, Sangdon Park, Susmit Jha†, Anirban Roy†, Oleg Sokolsky, Insup Lee (†SRI International)
➥ **ICML 2022** workshop on Principles of Distribution Shift,
➥ International Conference on Cyber-Physical Systems **(ICCPS) 2023** (Acceptance: 25.6%),
➥ **Best paper award nomination** at ICCPS 2023.

[2022A] Improving Neural Network Robustness via Persistency of Excitation
Kaustubh Sridhar, Oleg Sokolsky, Insup Lee, James Weimer
➥ American Control Conference (**ACC**) **2022**.

[2021B] Real-Time Detectors for Digital and Physical Adversarial Inputs to Perception Systems
Yiannis Kantaros, Taylor Carpenter, **Kaustubh Sridhar**, Yahan Yang, Insup Lee, James Weimer
➥ International Conference on Cyber-Physical Systems (**ICCPS**) **2021** (Acceptance rate: 26%).

[2021A] Real-Time Data-Predictive Attack-Recovery for Complex Cyber-Physical Systems
Lin Zhang, **Kaustubh Sridhar**, Mengyu Liu, Pengyuan Lu, F. Kong, Oleg Sokolsky, Insup Lee
➥ IEEE Real-Time and Embedded Technology and Applications Symposium (**RTAS**).

[2019] Finite-time, Event-triggered Tracking Control of Quadrotors
Kaustubh Sridhar, Srikant Sukumar
➥ Conference on Guidance, Navigation and Control (**EuroGNC**) **2019**.

Awards

2026 Charles Hallac and Sarah Keil Wolf Award for **Best ESE Dissertation**

2025 **Oral Presentation** (**top 1.8% of 11672 submissions**) for REGENT [2024B] at ICLR 2025

2023 **Best Paper Award Nomination** for CODiT [2022B] at ICCPS 2023

2022 **Top Reviewer (top 10%)**, NeurIPS 2022

2022 **Outstanding Reviewer (top 10%)**, ICML 2022

2023 **NSF Travel Grant**, International Conference on Cyber-Physical Systems (ICCPS) 2023

2022 **Student Travel Grant**, American Control Conference 2022

2019 **The Dean's Fellowship**, University of Pennsylvania

2019 **The Howard Bradwell Fellowship**, University of Pennsylvania

2018 **SN Bose Scholarship**, Govt. of India and the Indo-U.S. Science and Technology Forum

2015 **KVPY Fellowship**, Govt. of India

Selected Talks

Apr 2025 **Oral Talk at ICLR 2025** on REGENT, our generalist embodied agent [recording+Q&A].

2024-2025 **Training Adaptive and Sample-Efficient Generalist Embodied Agents.**
- [Apr 2025] NTU Singapore
- [Nov 2024] Google Deepmind
- [Nov 2024] Apple MLR

2023 **Learning Better Policies and Dynamics Models with Memory-Consistent and Memory-Constrained Neural Networks.**
- University of Pennsylvania (GRASP Lab) [video]
- University of Pennsylvania (Perception Action Learning Group)

2023 **Guaranteed Conformance of Neurosymbolic Generative (Dynamics) Models to Physics and Medical Constraints**
- Johns Hopkins University (CISS Session on Learning for Optimization and Control)
- Amazon Science (Deep Earth Reading Group)
- University of Pennsylvania (Formal Methods and Machine Learning Reading Group)

Press Coverage

2026 SIMA 2 Media Coverage: The Verge, MIT Tech Review, TechCrunch, The RunDown AI, The Decoder, WinBuzzer, SiliconANGLE, Forward Future, The Tech Buzz, Decrypt

2023 Making Better Decisions with AI, Penn Today (USA).

Service and Mentorship

2022 - Present **Reviewer**
ICLR 2026, 2025, 2024, ICML 2025, 2024, 2023, 2022, NeurIPS 2026, 2025, 2024, 2023, 2022, L4DC 2023, ICCPS 2025, 2022, CoRL 2026, 2025, RSS 2025

2020 - 2021 **Organizer**, *Reading Group in Robust Deep Learning*, University of Pennsylvania

2018 - 2019 **Team Lead**, *Department Academic Mentorship Program*, IIT Bombay
Led a team of 22 senior mentors to counsel 89 sophomores, 29 under-performing students.

Earlier Work Experience

May - Jul 2018 **Duke University**, *Undergraduate Summer Research Fellow*, Durham, NC.
Advised by Prof Miroslav Pajic, Cyber-Physical Systems Lab
- Developed a self-driving platform for intrusion detection testing [videos].

Jan - Dec 2018 **Indian Institute of Technology Bombay**, *Undergraduate Research Assistant*, India.
Advised by Prof Srikant Sukumar,
- Bachelor's thesis on real-time quadrotor control [2019].

Technical skills

Languages Python, C, C++
Machine Learning Pytorch, JAX, OpenAI Gym, Tensorflow, CUDA, Sklearn, Pandas
Robotics Mujoco, Bullet, CARLA, ROS, Gazebo

Key Coursework

Graduate Deep Learning, Reinforcement Learning, Convex Optimization, Probability, Computer Aided Verification
Undergraduate Data Structures and Algorithms, Linear and Nonlinear Control Theory, Adaptive and Optimal Control

Other Projects

Apr - May 2023 "Fixing Reward Hacking with Large Language Models."
- Created a framework for an RL agent in Deepmind AI Safety environments to leverage GPT4 to detect reward hacking, fix its own reward function, and adapt quickly to the new reward.

Teaching

Spring 22, Fall 24 **Teaching Assistant**, *CIS 541: Embedded Software for Life-Critical Systems*, UPenn
Spring 2021 **Teaching Assistant**, *CIT 595: Computer Systems Programming*, University of Pennsylvania